STEREOCHEMISTRY

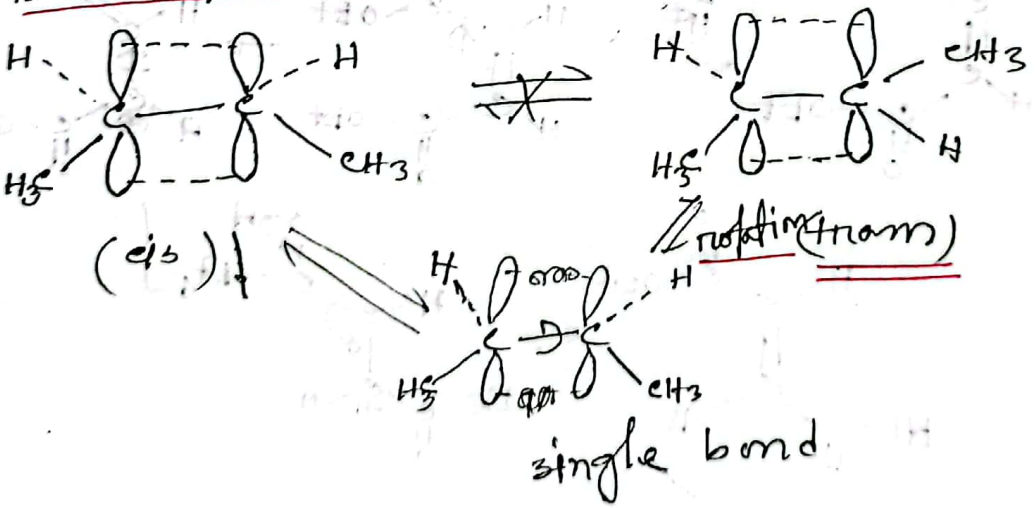
2017 | 2015

1 a) Name the methods used for inter conversion of cis-trans isomers.

Answer:

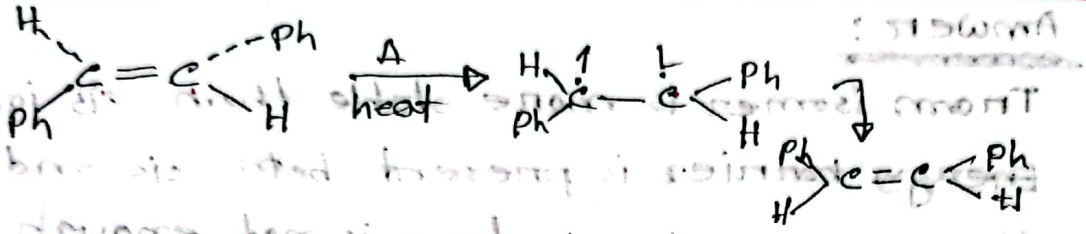
Trans isomer is more stable than cis isomer. Energy barrier is present betn cis and trans isomer. Room temperature is not enough for this inter-conversion and external energy is needed. There are two ways for interconversion ① by heating ② by UV-radiation.

In this interconversion, at first double bond break into single bond and rotation occurs in single bond, By 90° free rotation. cis isomer is converted into trans isomer.



(i) By application of heat :-

By heating the π electron in double bond break into free radical and rotation occurs and one isomer is converted into other.

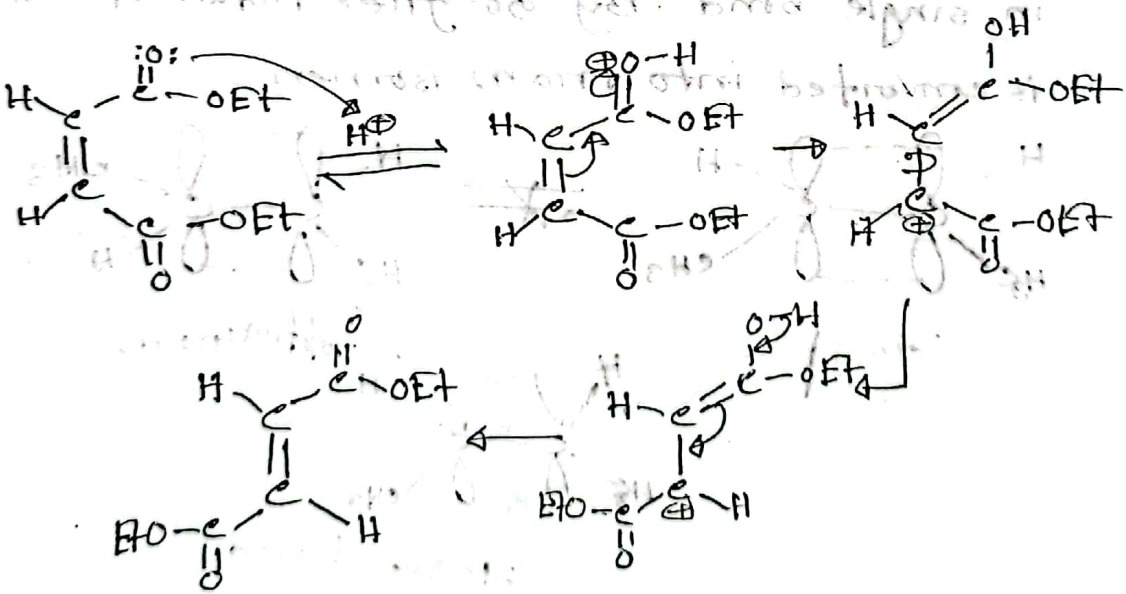


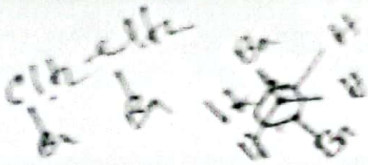
(ii) By irradiation with UV light :-

By UV light interconversion occur.



(iii) Interconversion of cis-ester into trans ester by acid catalyzed :-



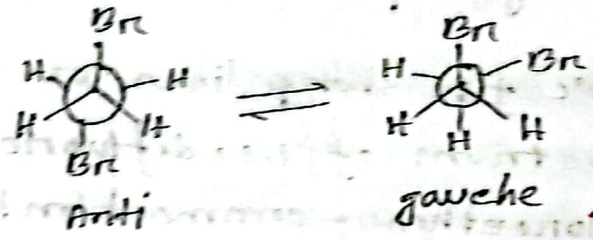


(10) What conclusion you can draw from the fact that extra lines in the IR spectrum of liquid 1,2 dibromoethane are split.

Answer:

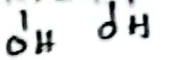
1,2 dibromoethane show extra lines in the IR spectrum of Liquid or gaseous state not in the solid state.

Because in solid state only one conformation present which is anti-form. But in liquid state two conformation present one is anti and one is gauche form. These conformation show extra lines.

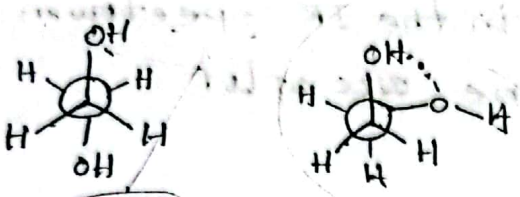


(11) A dil. solution in CCl_4 of ethylene glycol shows the following IR bands 3649 cm^{-1}, 3611 cm^{-1}.

Answer: The structure of ethylene glycol-



The conformation of ethylene glycol –



Anti

$\nu_{O-H} = 3694\,cm^{-1}$

free OH

gauche

$\nu_{O-H} = 3612\,cm^{-1}$

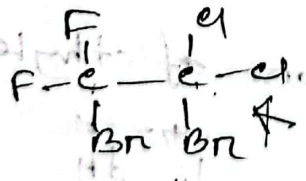
H-bonded

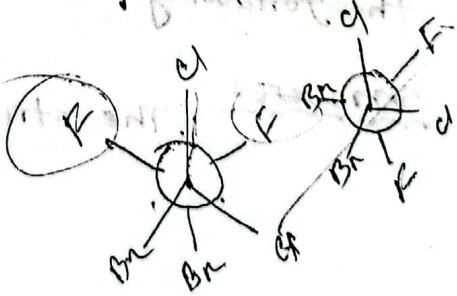
Two conformation is present. In anti conformation –O–H group is free and $\nu_{O-H} = 3694\,cm^{-1}$. But in gauche conformation two O–H group close to each other and intramolecular H-bond is formed betn two– OH group and $\nu_{O-H} = 3612\,cm^{-1}$.

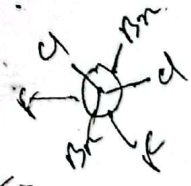
(c) At –80°C five sharp lines are shown in the nmr spectrum of 1,1-difluoro-1,2 dibromo-2,2 dichloroethane – comment on this observation.

Answer:

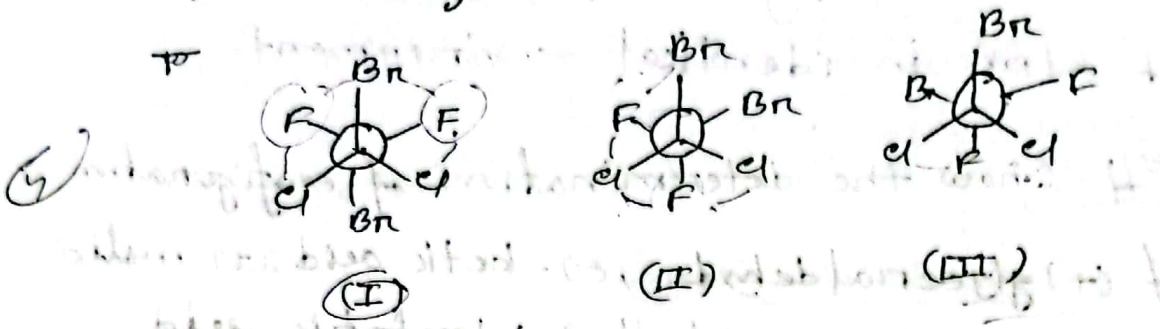
The structure of the compound –







Three conformation is possible, I, II and III.



(I) (II) (III)

These conformation present is proved by ^{19}F NMR.

At Room-temperature, ^{19}F nmr - given only one signal, which indicate it is probably from conformation-I, where two F atom contain same environment.

At -30°c temp. the broader line found.

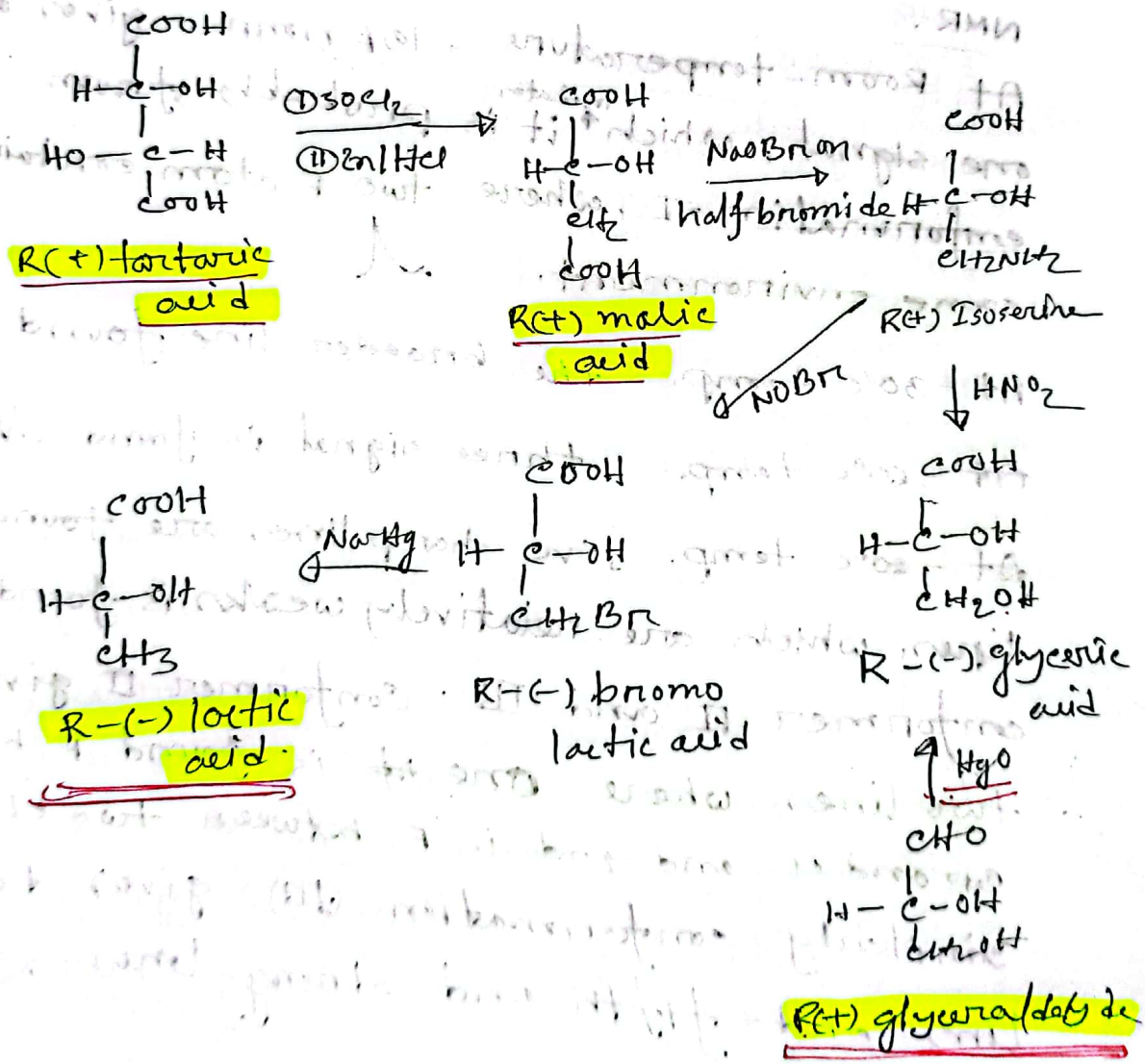
At -60°c temp. three signal is found.

At -80°c temp. five sharp liner are found. 4 liner which are relatively weaker is found by conformer II and III. Conformer II gives two liner where one is found F betn Br and Cl and 2nd is F between two Cl atom. similarly conformation (III) given two lins and fifth and strong liner are

found by conformation (II) in which two C atom in identical environment.

4. (ii) Show the determination of configuration of (+) glyceraldehyde, (-) lactic acid (or) malic acid with respect to (+) tartaric acid.

Answer:



two

Description:

① Here, reaction is started from R(+) tartaric acid, where in c-3, -OH group in the left side and in c-2, -OH group present in the right side, then c-3 contain OH group converted into -CH3 group without disturbed the bond betn c-2 and c-3 and give R(+) malic acid configuration also same.

(ii) R(+) malic acid converted to R(+) Isoserine, no changes the configuration because chiral c is not disturbed.

(iii) R(+) Iso-serine changes to glyceric acid- where configuration also same R(-) and R(+) glyceraldehyde gives R(-) glyceric acid, configuration also same, because chiral c is not effected any bond breaking or making, Both compound contain -OH group in the right side.

(iv) R(+) Iso-serine also changes to R(-) bromo lactic acid and by reduced bromolactic acid found R(-) lactic acid. Here, chiral

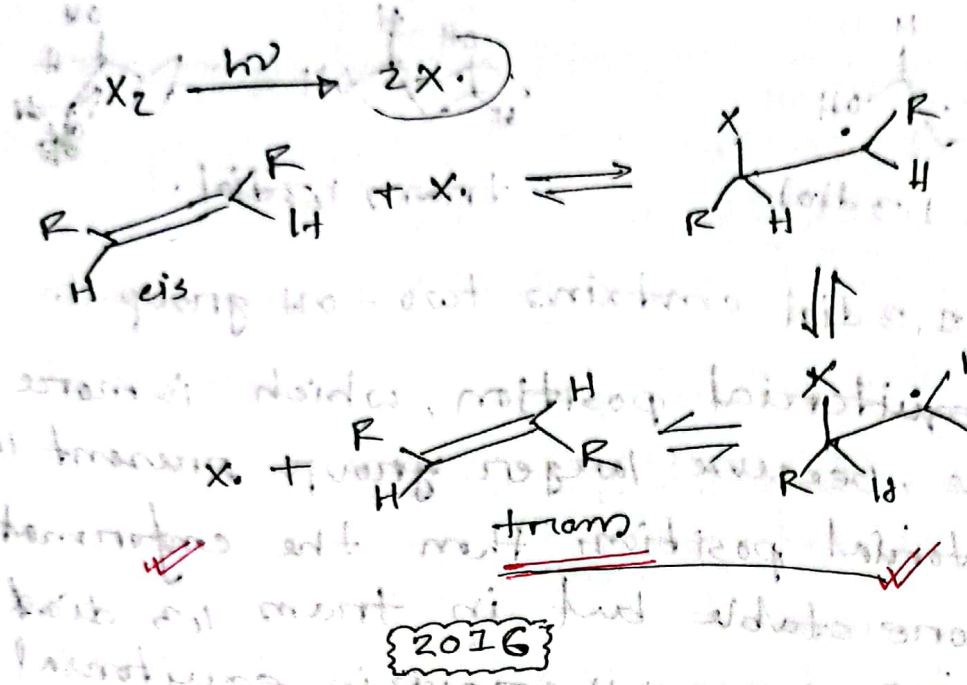
center is not disturbed and in e-2 -OH groups present in the right side. So (+)-glyceraldehyde, (-)-lactic acid, (+) malic acid contain -OH group in e-2 in the right side like (+)-tartaric acid.

2. (iv) Halogens are used for stereomutation of geometrical isomers.

Answer:

The interconversion of geometrical isomer is also called stereomutation. The cis isomer is converted to trans isomer. In this interconversion extra is not occurs in the RT, Extra energy is needed for this interconversion. Double bond converted into single bond then rotation occurs and new isomer formed.

Halogen are used for this inter-conversion At first halogen converted to radical and then conversion occurs.

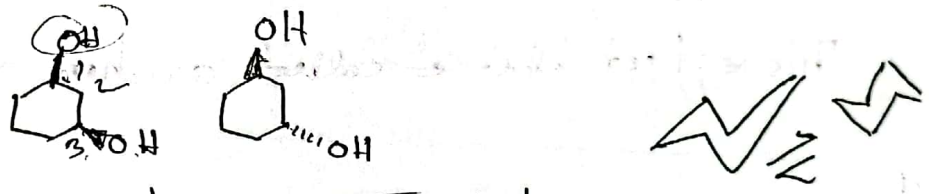


2016

6.(vii)

cis-Cyclohexane-1,3 diol is more stable than trans form.

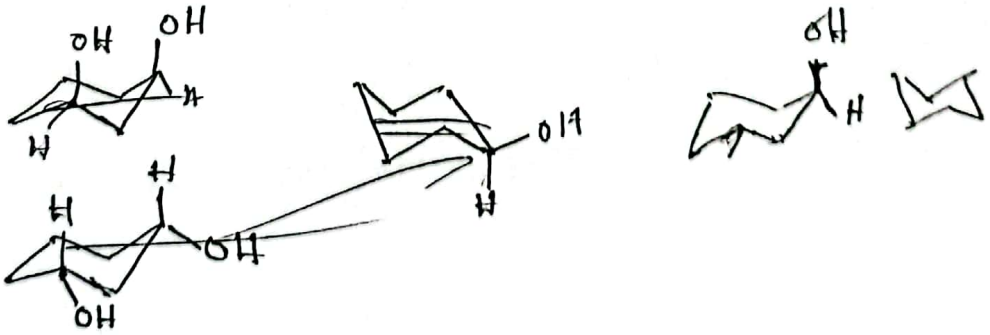
Answer:

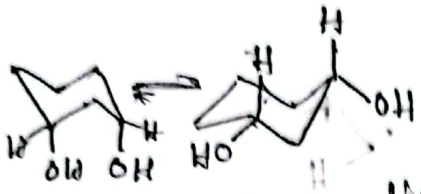


cis-diol trans diol.

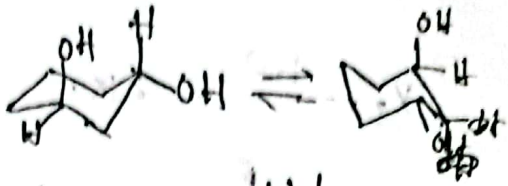
cis-diol is more stable than trans diol.

The conformation structure of this diol—





cis 1,3 diol



trans 1,3 diol.

cis 1,3 diol contains two -OH group in the equitorial position, which is more stable, because larger group present in equitorial position than the conformation is more stable but in trans 1,3 diol contain one -OH group in equitorial position and one axial position which is less stable than cis 1,3 diol.

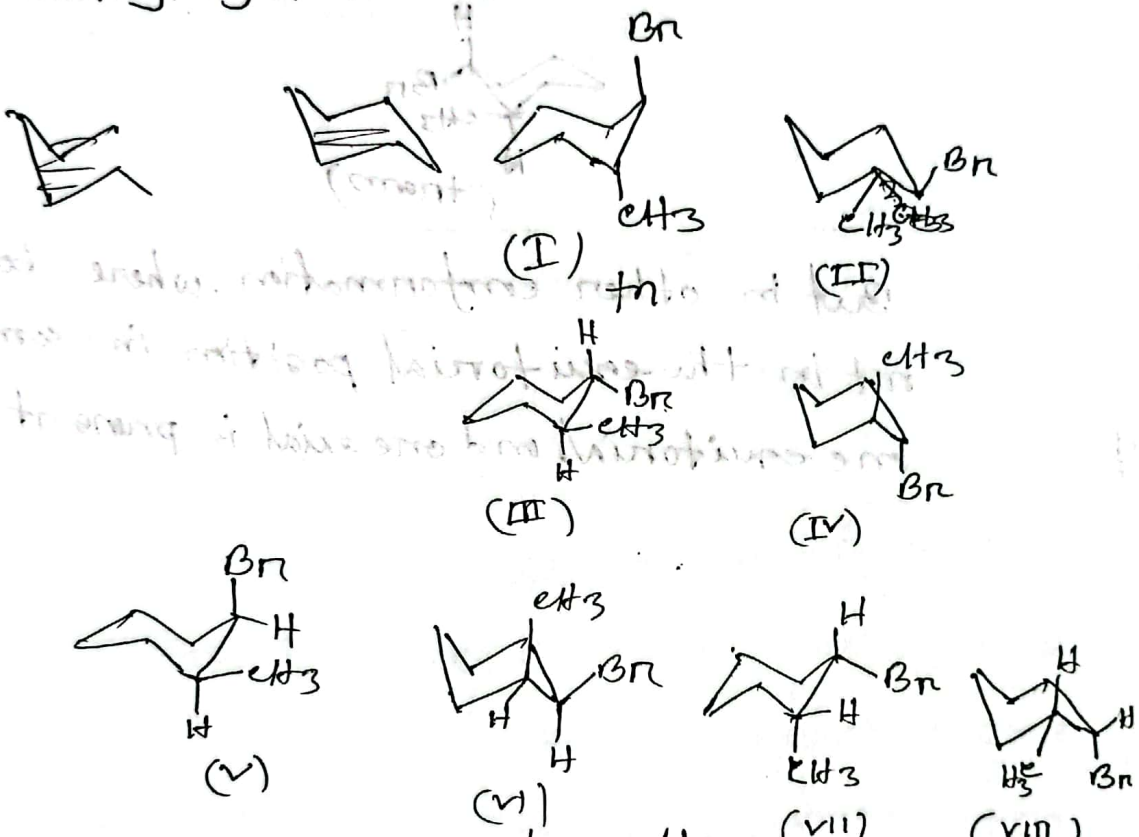
This effect also called anchor effect.

2014

7. a) Draw the different conformations of 1-Bromo-2 methyl cyclohexane, hence find the most stable conformation of the above identifying cis and trans conformation

Answer:

The different conformation of 1-bromo-2-methyl cyclohexane -

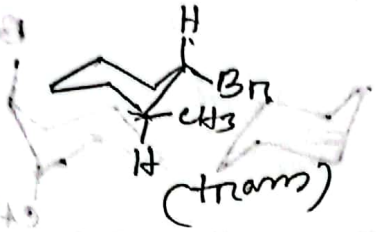


Here, 8 conf chair conformation present.

In this conformation (I-IV) is trans form and (V-VIII) is cis form.



The most stable conformation is trans form, where both group (Br and $-CH_3$) present in equitorial position.



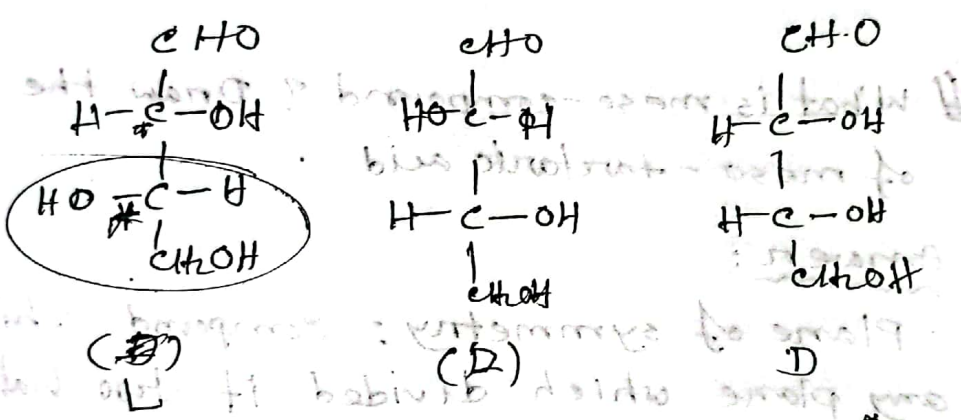
But in other conformation, where both are not in the equitorial position in same time, one equitorial and one axial is present.



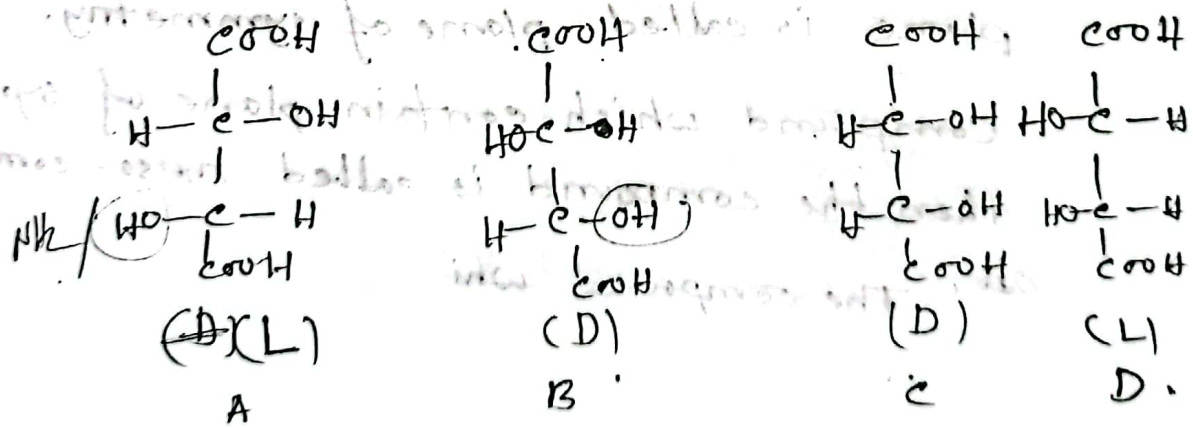
Q11 Differentiate between D and L ~~and~~ enantiomer and diastereomer.

Answer:

When -OH / -NH_2 group in the lower chiral carbon contain in the right-side, then this configuration is called D and when this group on the left side, then this configuration called L.



Consider a example, $COOH-CH(OH)-CH(OH)-COOH$.



The compound A and B are called enantiomer of each other because, mirror image of each other and non superimmpose each other and -OH group present left side which is L and in the right side which is D.

The ~~compound~~ A - pair A and C, B and D is also called diastereomer. where ~~C = D~~ they are not mirror image of each other and non superimpossible.

✓ Q) What is meso-compound? Draw the structure of meso-tartaric acid.

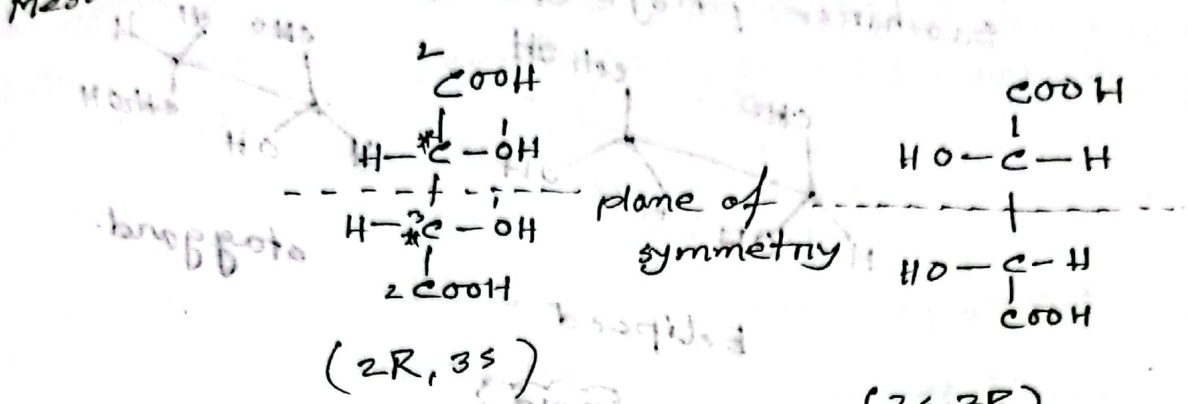
Answer:

Plane of symmetry: compound which contain any plane which divided it two halves, and one* is superimmpose to another, then this plane is called plane of symmetry.

Compound which contain plane of symmetry then the compound is called meso-compound.

Or, ~~The compound whi~~

Meso-tartaric acid –



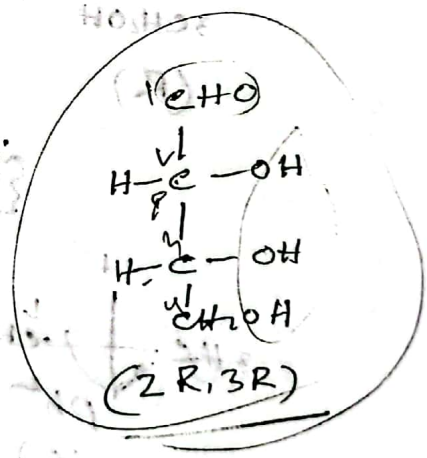
(2R, 3S) (2S, 3R)

Meso-tartaric acid

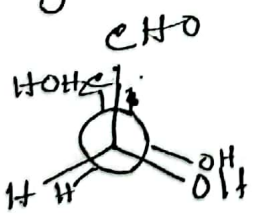
Q// Draw the structure of D-Erythrose or (2R, 3R) trihydroxybutanal in sawhorse, Newman and Fischer projection forms.

Answer:

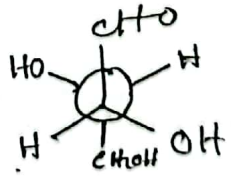
Fisher projection:



Newman projection:

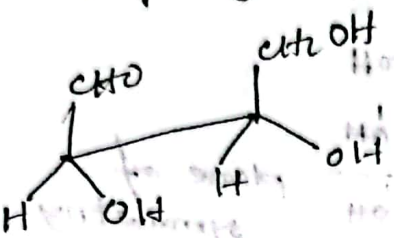


Eclipsed

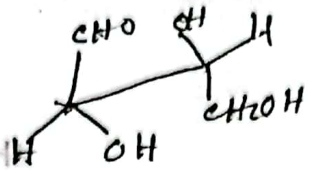


Staggard.

Sawhorse projection:



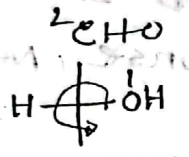
Eclipsed



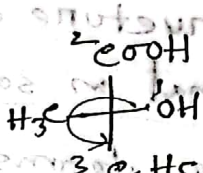
staggard.

2013

Q/ Specify as R or S the following compounds.

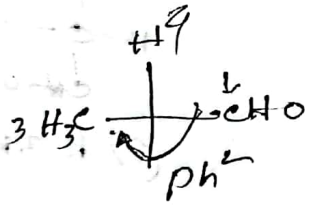


(R)



(R)

2012



(S)

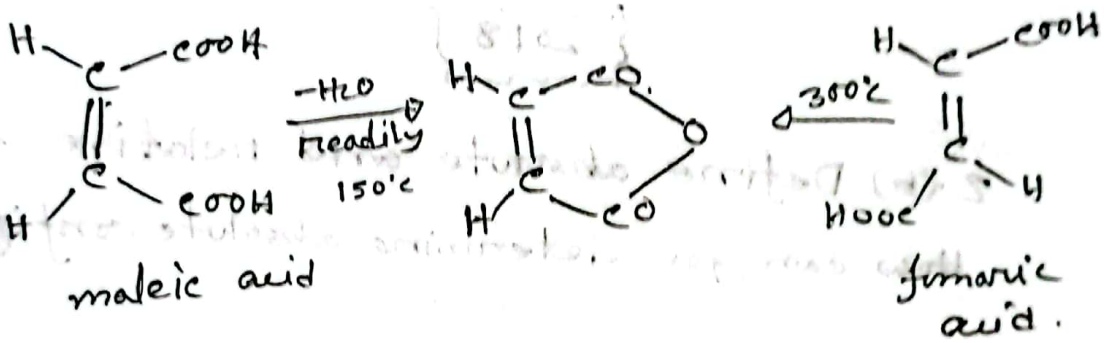
(R)

{2012}

6. a) Discuss the method of cyclization for the determination of configuration of geometrical isomers.

Answer:

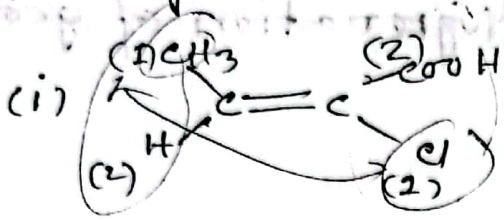
By cyclization we determine the configuration of geometrical isomer. When two groups are more closely present then intramolecular reaction are occur. maleic acid and fumaric acid, only one readily form cyclic anhydride when two groups are so close



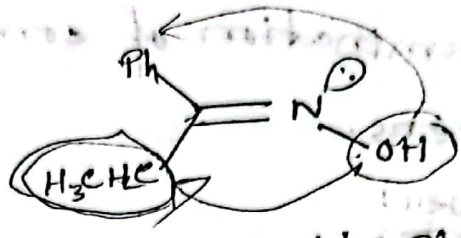
In fumaric acid, two cooH groups are opposite side and more head are needed for formation of cyclic anhydride.

So we called that the cis form show cyclization readily.

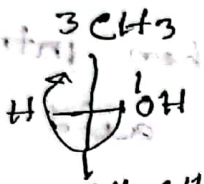
d// Assign nomenclature for the followings:

(i) 

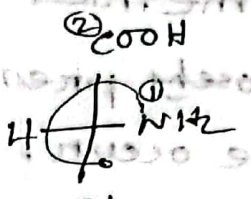
(E)

(ii) 

Anti (Ph) on syn ethyl



(S)



(R)

2018

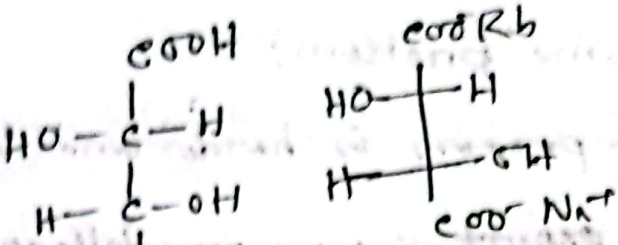
2.(b) Define absolute and relative configuration. How can you determine absolute configuration.

Answer:

Absolute configuration:-

In three dimentional arrangement of a molecule, the actual arrangement of different atom or groups that is determined by x-ray, electron diffraction is called absolute configuration.

condition: must be single crystal.

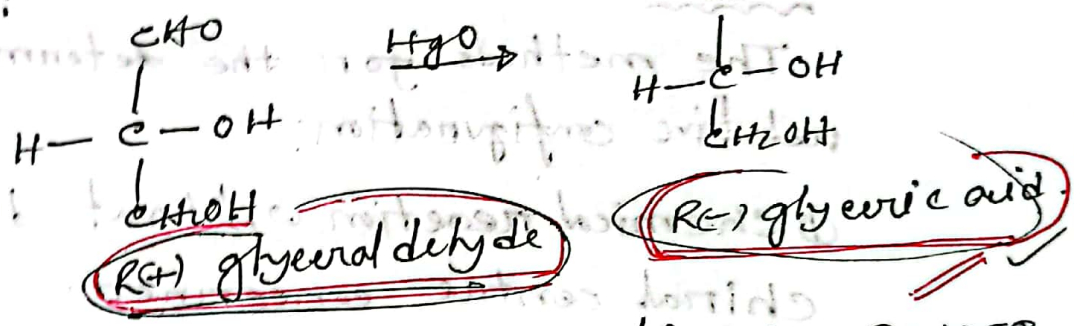


(+) tartaric acid. Na/Rb salt of tartaric acid

Relative configuration :-

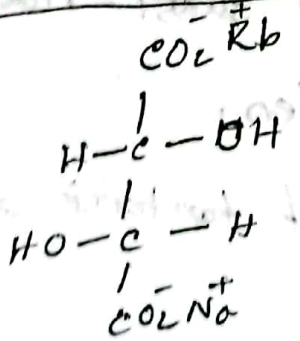
The configuration of a molecule, that is determined by comparing with standard configuration.

Example :



(R+) glyceraldehyde → (R+) glyceric acid

Determination of absolute configuration :- In 1952 Bijvot experiment determine the absolute confi. By - x-ray diffraction determine this configuration. of Na, Rb salt of tartaric acid -



There are some problem:-

(I) Ab. This process is harder and lengthy process.

(II) Needed expert x-ray crystallographer.

Slide 4

Q Name the methods for the determination of relative configuration. Discuss the co-relation method of chemical interconversion not affecting bonds to the asymmetric atom.

Answer:

The methods for the determination of relative configuration:-

(I) Chemical reaction without displacement at chiral centre concerned.

(II) Chemical reaction with displacement of chiral centre concerned.

(III) X-ray analysis.

(IV) Optical rotations.

(V) The study of quasi-racemic compound.

(VI) Enzyme studies.

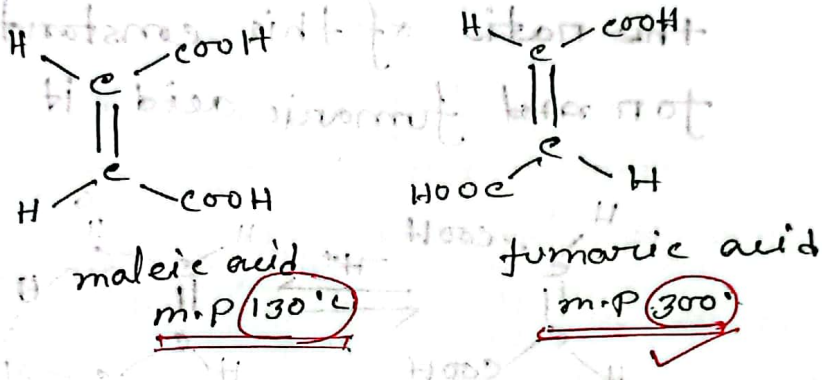
Then show reaction of tartaric acid)

lecture:

Melting point of cis- trans isomer ...

The melting point of trans isomer is greater than cis-isomer. Because ; in trans isomer more symmetry is present, than cis-isomer. so, the crystal lattice is more fit and more heat energy is needed for melting this trans isomer.

Example:

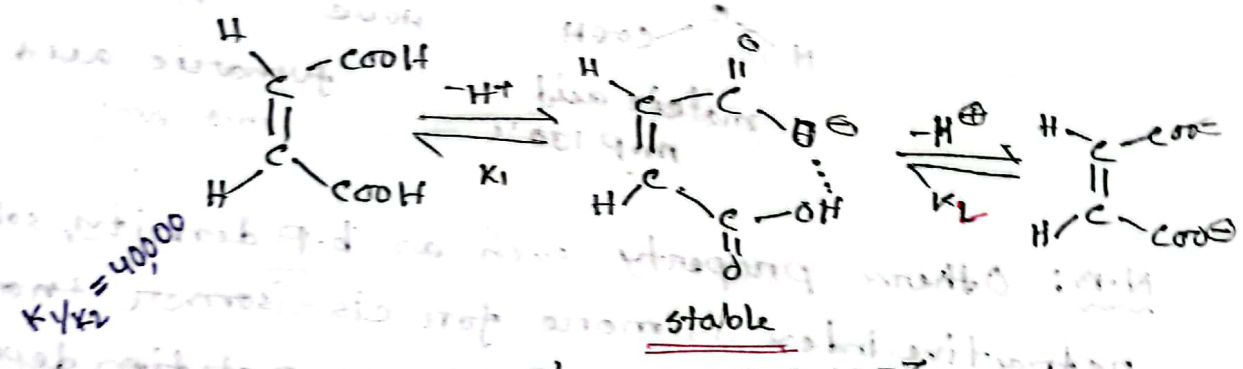


N.B: Others property such as b.P density, solubilit refractive index is more for cis-isomer than trans isomer because these properties depends on volume. Trans isomer is has more volume and there properties are lower for trans isomer than cis-isomer.

Acidic strength of cis-trans isomers.

Answer:

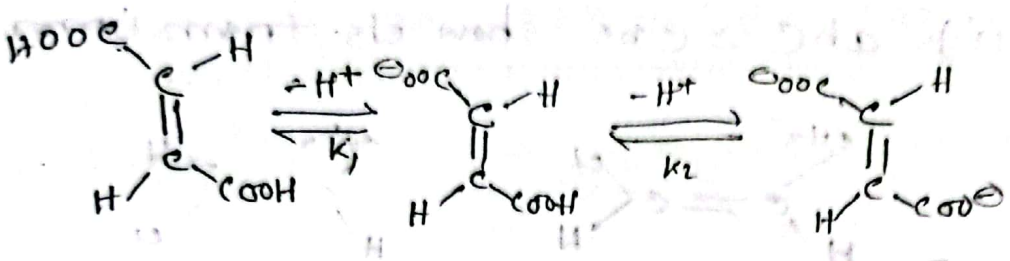
cis-acid is more acidic than trans acid. Because in cis-acid, both -COOH group present in the same side and easily one H^+ is release and anion is stablized by forming H bond. The first acid constant K_1 is higher than K_2 and ratio K_1/K_2 is high. For maleic acid the ratio of this constant is 40,000 and for fumaric acid, it is 32.



$K_1 = 1.38 \times 10^{-2}$ $K_2 = 3.2 \times 10^{-7}$.

$\frac{K_1}{K_2} = 40,000$.

On the other hand, in trans isomer -COOH groups present opposite side and K_1 and K_2 is has smaller difference.



$K_1 = 1.0 \times 10^{-3}$ $\quad K_2 = 3.2 \times 10^{-5}$.

$K_1/K_2 = 32$

$$\frac{K_1}{K_2} = 32.$$

So, cis acid is more acidic than trans acid.

Condition for cis-trans isomer.

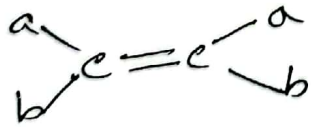
Answer:

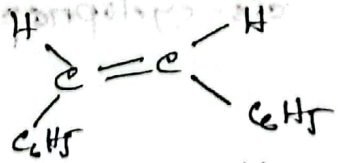
(a) Double bond must be present. Single bond easily rotate but in double bond, π bond present, so free rotation is not possible. Because of the rigidity double bond show isomer.



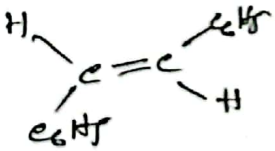
(b) The end carbon of the double bond, must be present different group or atom.—

① abc=cab, show cis-trans isomer.



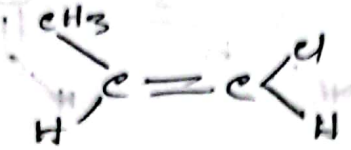


cis-stilbene

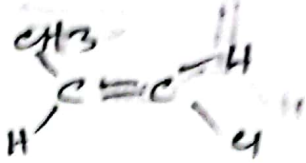


trans.

(ii) abC=Cbc show cis-trans isomer.

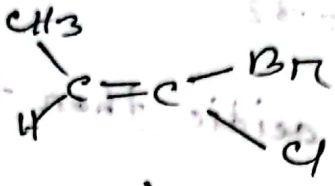


cis

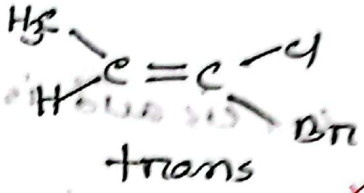


trans

(iii) abc=ccd will show cis-trans isomer



cis



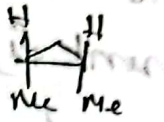
trans

(c) energy barrier between cis-trans isomer is higher.

** Cyclic compound also show cis-trans isomer, because in cyclic compound c-c bond can't show free rotation like alkene and show cis-trans isomer.



cis-cyclopropane



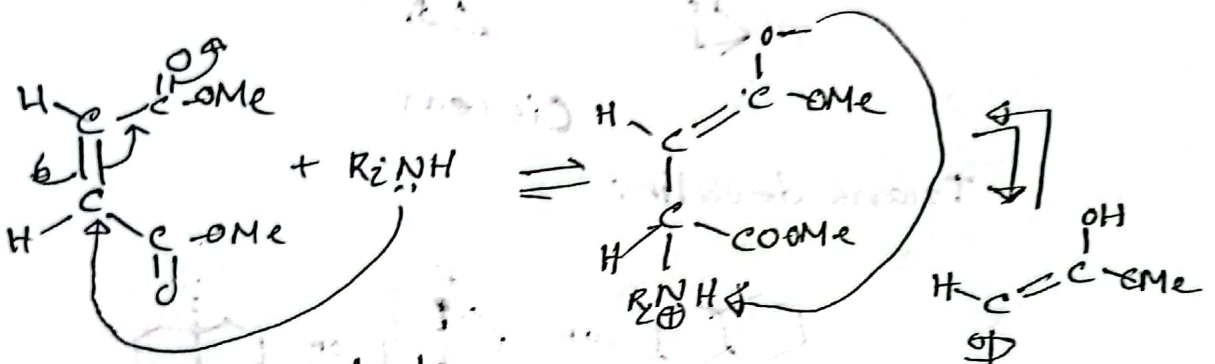
trans

2018

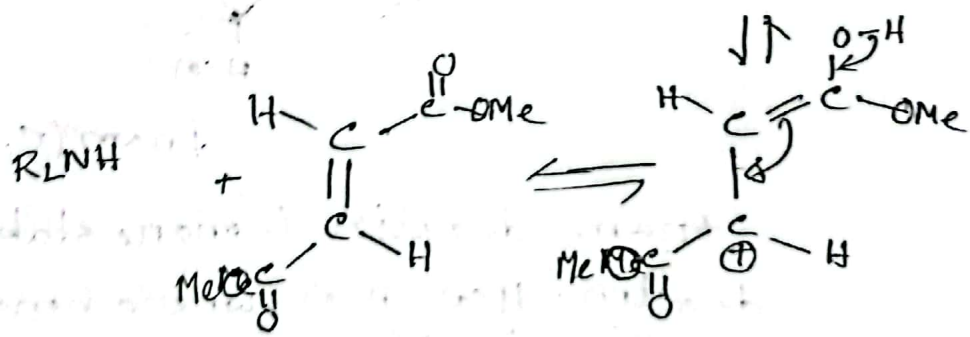
6. (v) Dimethyl maleate is isomerized to dimethyl fumarate by means of secondary amines.

Answer:

Interconversion is occured between cis-trans isomer by base catalyzed and secondary amine also used for this interconversion—



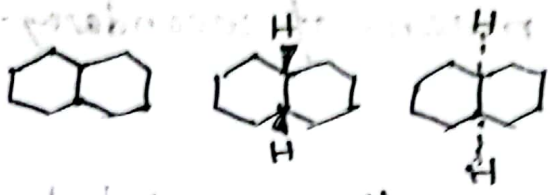
Dimethyl maleate



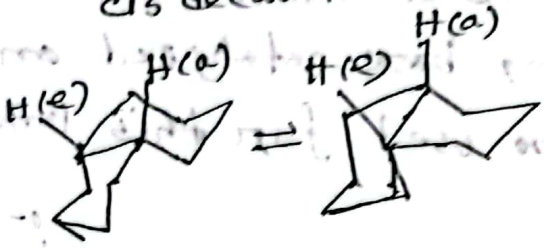
Dimethyl fumerate

20-08-2
Must be read the topic.

Bi-cyclic compound :-

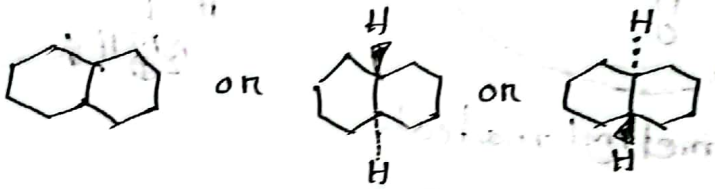


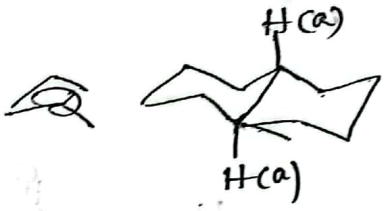
cis decalin



Cis (ea)

Trans decalin:





tram (e,e)

tram decalin is more stable than cis decalin. H(a) – H(a) ; where bond angle 180°. filping is not occured in tram. But cis-decalin easily flipping to another.

THE END
MOSHEUR
CHEM-96

Monjur Rahman